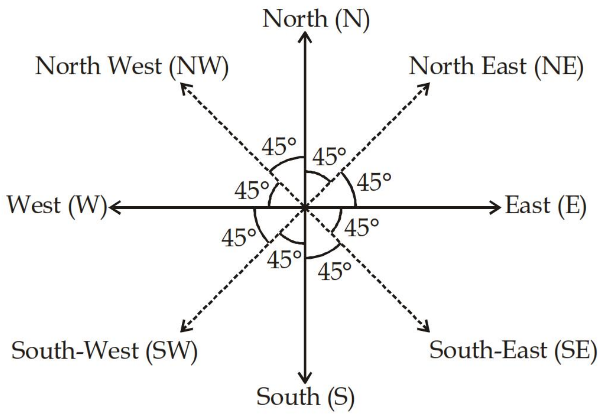
Chapter 11

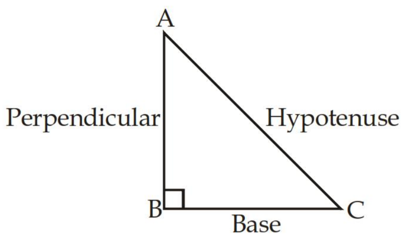
Direction and Distance

It is introduced in reasoning tests to gauge the 'sense of direction' of the candidate. But as the reasoning tests have become frequent in competitive examinations, the usage of such question has been increased. Today, direction tests are not only used in reasoning tests for checking 'sense of direction' but also for logical comprehension of particular situations

- There are four main directions : -
  East, West, North and South
- There are four cardinal directions :-
  North-East (N-E), North-West (N-W)
  South-West (S-W) & South-East (S-E)



- One should be aware of basic geometric concepts, to find the shortest distance, such as Pythagoras Theorem.



$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$

$AC^2 = BC^2 + AB^2$

$\boxed{AC = \sqrt{BC^2 + AB^2}}$

Solved Examples

TYPE – 1 :

1. Ramu moves 3 km towards the North and turns to right upto 4 km. What is the distance from his starting point ?

(a) 10 cm (b) 8 km

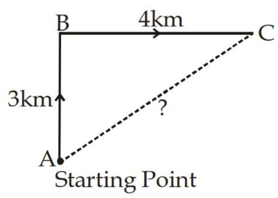
(c) 5 km (d) 14 km

Sol. $AC^2 = AB^2 + BC^2$

$AC^2 = 3^2 + 4^2$

$AC^2 = 9 + 16$

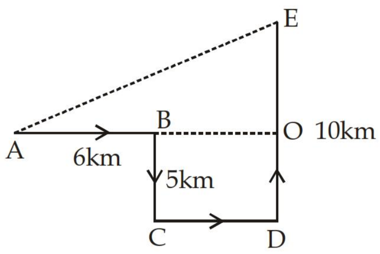
$AC = \sqrt{25} = 5$ km



$\therefore$ Distance from his starting point = 5 km.

2. A man walks 6km to the east and then turns to the south and walks 5 km. Again he turns to the east and walks 6km. Next, he turns northwards and walks 10km. How for is he now from his starting point ?
(a) 15 km (b) 13 km
(c) 7 km (d) 12 km

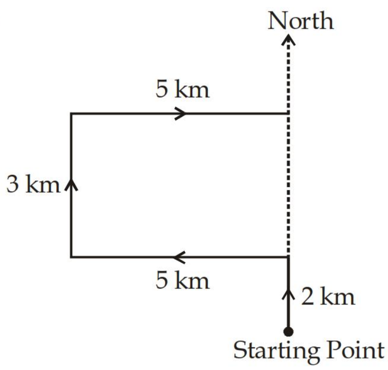
Sol.



BO = CD = 6 km
BC = OD = 5 km
$\therefore$ EO = ED – OD = (10 – 5) km = 5km
AO = AB + BO = 6 + 6 = 12km
$\therefore$ $AE^2 = AO^2 + OE^2$
$AE^2 = 12^2 + 5^2$
$AE^2 = 144 + 25$
$AE = \sqrt{169} = 13$km

3. Rahul walks 2km Northward and takes a left turn walks 5km and then turns right, walks 3km and again turning right, walks 5km. In which direction is he now from the starting point?
(a) East (b) North
(c) West (d) South

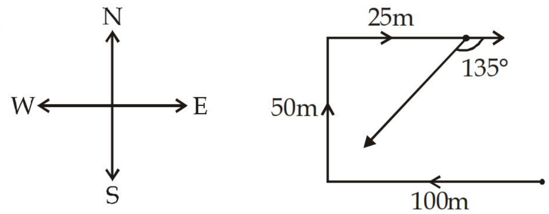
Sol.



Answer is North

4. A person is facing North, he turns to his left and moves 100 m. Then he turns his right twice and goes straight 50m and 25m respectively. Then he turns to right up to 135° and starts moving. Now in which direction he is moving ?
(a) South-East (b) South-West
(c) South (d) West

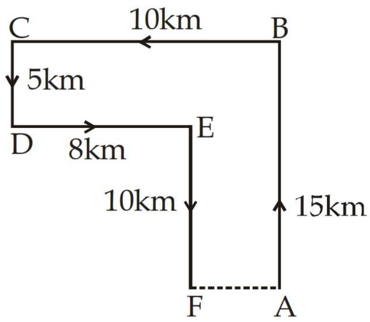
Sol.



He is moving towards South-West.

5. Raju left for his office in his car. He drove 15 km towards North and then 10km towards west. He then turned to the south and covered 5km. Further he turned to the east and moved 8km. Finally, he turned right and drove 10 km. How far and in which direction is he from his starting point ?
(a) 2 km west (b) 2 km East
(c) 3 km North (d) 6 km North

Sol.



Raju's distance from the starting point A
= AF = (BC – DE) = (10 – 8) km = 2 km
$\therefore$ Answer is ; 2km west.

Practice Set

1. If A is to the south of B and C is to the east of B, in which direction is A with respect to C?
(a) North-east (b) North-west
(c) South-east (d) South-west

2. A is 40 m South-west of B. C is 40 m South-east of B. Then, C is in which direction of A?
(a) East (b) West
(c) North-east (d) South

3. Raj travelled from a point X straight towards east to Y at a distance of 80 metres. He turned right and walked 50 metres, then again turned right and walked 70 metres. Finally, he turned right and walked 50 metres. How far is he from the starting point?
(a) 10 metres (b) 20 metres
(c) 50 metres (d) 70 metres

4. Aditya went 15 kms to the west from his house, then turned left and walked 20 kms. He then turned East and walked 25 kms and finally turning left covered 20 kms. How far was he from his house?

(a) 5 kms (b) 10 kms

(c) 40 kms (d) 80 kms

5. From his house, Lokesh went 15 kms to the North. Then he turned West and covered 10 kms. Then, he turned South and covered 5 kms. Finally, turning to East, he covered 10 kms. In which direction is from his house?

(a) East (b) West

(c) North (d) South

6. 'Z' started walking straight towards South. He walked a distance of 15 metres and then took a left turn and walked a distance of 30 metres. Then he took a right turn and walked a distance of 15 metres again. Z is facing which direction ?

(a) North East (b) South

(c) North (d) South-West

7. Alok walked 30 metres towards east and took a right turn and walked 40 metres. He again took a right turn and walked 50 metres. Towards which direction is he from his starting point?

(a) South (b) West

(c) South-West (d) South-East

8. A man is facing north-west. He turns 90° in the clockwise direction and then 135° in the anticlockwise direction. Which direction is he facing now?

(a) East (b) West

(c) North (d) South

9. A man is facing north-west. He turns 90° in the clockwise direction, then 180° in the anticlockwise direction and then another 90° in the same direction. Which direction is he facing now?

(a) South (b) South-west

(c) West (d) South-east

10. I am facing east. I turn 100° in the clockwise dircrtion and then 145° in the anticlockwise direction. Which direction am I facing now?

(a) East (b) North-East

(c) North (d) South-West

11. Kishankant walks 10 km towards North. From there, he walks 6km towards South. Then he walks 3km towards East. How far and in which direction is he with reference to his starting point ?

(a) 5km West (b) 5km North-East

(c) 7km East (d) 7km West

12. Gaurav walks 20 metres towards North. He then turns left and walks 40 metres. He again turns left and walks 20 metres. Further, he moves 20 metres after turning to the right. How far is he from his original position?

(a) 20 metres (b) 30 metres

(c) 50 metres (d) 60 metres

13. Radha moves towards South-east a distance of 7 km, then she moves towards West and travels a distance of 14 m. From here, she moves towards North-west a distance of 7 m and finally she moves a distance of 4 m towards East and stood at that point. How far is the starting point from where she stood?

(a) 3 m (b) 4 m

(c) 10 m (d) 11 m

14. A rat runs 20' towards East and turns to right, runs 10' and turns to right, runs 9' and again turns to left, runs 5' and then turns to left, runs 12' and finally turns to left and runs 6'. Now which direction is the rat facing ?

(a) East (b) West

(c) North (d) South

15. I am facing South. I turn right and walk 20 m. Then I turn right again and walk 10 m. Then I turn left and walk 10 m and then turning right walk 20 m. Then I turn right again and walk 60 m. In which direction am I from the starting point?

(a) North (b) North-West

(c) East (d) North-East

16. Rohit walked 25 metres towards South. Then he turned to his left and walked 20 metres. He then turned to his left and walked 25 metres. He again turned to his right and walked 15 metres. At what distance is he from the starting point and in which direction?

(a) 35 metres East (b) 35 metres North

(c) 40 metres East (d) 60 metres East

17. Starting from a point P, Sachin walked 20 metres towards South. He turned left and walket 30 metres. He then turned left and walked 20 metres. He again turned left and walked 40 metres and reached a point Q. How far and in which direction is the point Q from the point P ?

(a) 20 metres West (b) 10 metres East

(c) 10 metres West (d) 10 metres North

18. A man walks 1 km towards East and then he turns to South and walks 5 km. Again he turns to East and walks 2 km, after this he turns to North and walks 9 km. Now, how far is he from his starting point?

(a) 3 km (b) 4 km

(c) 5 km (d) 7 km

19. From his house, Lokesh went 15 kms to the North. Then he turned West and covered 10 kms. Then, he turned South and covered 5 kms. Finally, turning to East, he covered 10 kms. In which direction is he from his house?

(a) East (b) West
(c) North (d) South

20. Two buses start from the opposite point of a main road, 150 kms apart. The first bus runs for 25 kms and takes a right turn and then runs for 15 kms. It then turns left and runs for another 25 kms and takes the direction back to reach the main road. In the meantime, due to a minor breakdown, the other bus has run only 35 kms along the main road. What would be the distance between the two buses at this point ?

(a) 65 kms (b) 75 kms
(c) 80 kms (d) 85 kms

21. Sohan started from point X and travelled forward 8 km up to point Y, then turned towards right and travelled 5km up to point Z, then turned right and travelled 7km up to point A and then turned towards right and travelled 5km up to B. What is the distance between points B and X?

(a) 1 km (b) 2 km
(c) 3 km (d) 4 km

22. Raj starts from his office facing west and walks 100 m straight, then takes a right turn and walks 100 m. Further he takes a left turn and walks 50 m. In which direction is Raj now from the starting point?

(a) North-East (b) South-West
(c) West (d) North-West

23. From a point 'Q' Alok starts walking towards south and after walking 25 m, he turns to his left and walks 40 m and reaches point 'F'. In which direction is he with reference to the starting point 'Q'?

(a) South-West (b) Noth-East
(c) North-West (d) South-East

24. Ram is facing north. He proceeded straight for a distance of 10km, then he turned left and proceeded straight a distance of 15km. At last he turned left again and proceeded for another 10km. How far is he from the starting point ?

(a) 10km (b) 5km
(c) 12km (d) 15km

25. Mohan proceeded straight in the direction of north for a distance of 3 km from his house. Then he turned right and proceeded straight for a distance of 2km. Then he turned right again and proceeded straight for a distance of 5 km. Then he turned right twice and proceeded straight for a distance of 2-2km respectively.Which direction is he facing now?

(a) North (b) South
(c) West (d) East

26. Two men 'A' and 'B' Starts their' Journey from a special point 'X'. 'A' walks 4 km in the direction of west. 'B' walks 1 km in the direction of north then he turns left and walks straight 4 km. 'A' turns right and walks 2 km straight. How far is now 'A' from 'B'?

(a) 1 km (b) 2 km
(c) 3 km (d) 4 km

27. Morning time, Sohan goes towards sun for a distance of 50m. Then he turns right twice and goes straight 40m and 20m respectively. At last he turns once more right and goes straight for the ditance of 40m. How far is he now from his starting point ?

(a) 20 m (b) 30 m
(c) 50 m (d) 60 m

28. Rahul is facing west. Then he turned right three times and walked straight for a distance of 15 km, 15 km, and 35 km respectively, How far is he from starting point ?

(a) 15 km (b) 25 km
(c) 30 km (d) 20 km

Direction (29 – 31): Read the following information for answering the questions that follows :

In a playground, A, B, C, D, & E are standing as described below facing the North.

(1) B is 50m to the right of D.
(2) A is 60m to the South of B.
(3) C is 40m to the west of D.
(4) E is 80m to the north of A.

29. If a boy walks from C, meet D followed by B, A and then E, how many meters has he walked if he has travelled the straight distance all through?

(a) 120 (b) 150
(c) 170 (d) 230

30. What is the minimum distance (in metre approximately) between C and E?

(a) 53 (b) 78
(c) 92 (d) 120

31. Who is to the South-East of the person who is to the left of D?

(a) A (b) B
(c) C (d) E

32. A man is facing west direction. He turns twice in clockwise direction 45° and 180° respectively. He then turns 270° in the anticlockwise direction. which direction is facing now?

(a) South (b) North-East
(c) West (d) South-West

33. Four aeroplanes of Air Force viz, A, B, C, D, started for a demonstration flight towards east. After flying 50 km planes A and D flew towards right, planes B and C flew towards left. After 50km, planes B and C flew towards their left, planes A and D also towards their left. In which directions are the aeroplanes A, B, D, C repectively flying now?

(a) East, West, East, West
(b) West, East, West, East
(c) North, South, East, West
(d) South, North, West, East

Direction (34 – 35): There are 6 check-posts A, B, C, D, E and F. Check-post F is 15km to the north of D, which is 25 km to the North-East of B. Check post A is 5 km west of E and 15 km to the South-West of C. Check post B, A and E are in a straight line and A is to the east of B. The check post B and E are 30 km apart from each other.

34. If a jeep moves from E to F, via A, B and D, how much distance will it have to cover?
(a) 70 km (b) 120 km
(c) 100 km (d) 90 km

35. Which check post is to the South-West of D?
(a) A (b) B
(c) C (d) D

36. Vishesh goes in north direction for 5 km. Then he turns right and goes 5 km straight. Then he turns left and goes 10km straight. At last, he turned clockwise direction 45° and goes straight. In which direction is Vishesh going now?
(a) South (b) South-West
(c) North-East (d) North

37. Raju starts his journey in north-west direction 2 km straight. Then he turns 90° clockwise and walks 2 km straight. Then once more he turns 90° clockwise and walks 2km straight. In which direction is Raju from his starting point?
(a) West-South (b) North-East
(c) East-South (d) West

38. A watch reads 3 : 30 If the minute hand points 'North'. In which direction will the hour hand point?
(a) North-West (b) North
(c) West (d) East

39. A watch reads 9:00. If the minute hand points 'North' then minute hand turned 135° clockwise. Then what is the time shown by the watch and in which direction will the minute hand points ?
(a) South-East, 9:22:5 (b) South, 9:30
(c) West- North, 9 : 10 (d) South-West, 9:40

40. A child is looking for his mother. He went 80 meters in the East before turning to his right. He went 20 meters before turning to his right again to trace his mother at his married sister's house, 20 meters from this point. His mother was not there. From there he went 100 meters to his north where he met his mother who was shopping there in the market. How far did the son meet from the starting point?
(a) 80 m (b) 60 m
(c) 100 m (d) 140 m

41. Satish starts from A and walks 2 km east upto B and turns sothwards and walks 1 km upto C. At C he turns to east and walks 2 km upto D. He then turns northwards and walks 4 km to E. How far is he from his starting point?
(a) 3 km (b) 4 km
(c) 5 km (d) 6 km

42. A man starts his journey from a point 'X'. He goes in north direction for 5 km. Then he turns his left and goes 2 km, then he turns his right twice and goes straight 3 km and 2 km respectively. How far is he from his starting point?
(a) 7 km (b) 8 km
(c) 9 km (d) 6 km

43. A lady runs 12 km towards north then 6 km towards south and then 8 km east. How far is she from her starting point and in which direction?
(a) 5 km North-East (b) 5 km East
(c) 10 km North-East (d) 10 km West

44. Two men start walking from one point towards opposite direction. After walking 3 km straight the both turn right wards and walk straight for the distance of 4km. How far are they both from each-other?
(a) 8 km (b) 7 km
(c) 9 km (d) 10 km

45. Point 'K' is situated in noth-west direction from 'P' at the distance of 2 km. Point 'R' is situated in south-West direction from 'K' at the distance of 2 km. Point 'M' is situated in North-West direction form 'R' at the distance of 2km. At last 'T' is situated in South-West direction from 'M' at the distance of 2 km. In which direction is 'T' from point 'P'?
(a) East (b) West
(c) South (d) North

46. If South-East is called East, South-West is called South and in the same way directions are changing.Then what would we say the North direction?
(a) North-West (b) North-East
(c) South-West (d) South-East

47. On a crossing, there is symbol telling the direction of roads as North, South, East, West. The symbol turned due to an accident. The part which is telling north direction now tells the west direction. If a traveller goes to south direction with the help of turning symbol. Then really, in which direction is he going now?
(a) West (b) East
(c) North (d) None of there

Direction (48 – 50): Based on informations given below. Read carefully and choose best alternatives.

In a playground five friends A, B, C, D and E are standing facing north in order as described below.

(i) B is standing at the right of D at distance of 50 m.

(ii) A is standing in the south direction of B at 60 m distance.

(iii) 'C' is in the west of 'D' at the distance of 40 m.

(iv) 'E' is in the north of A at 80 m distance

48. If a boy, who stands near 'C' starts walking and reach to 'D' then B, A, E respectively.

How many km he walk from C to E?

(a) 120 m (b) 150 m
(c) 170 m (d) 230 m

49. How many distance is between C and E?

(a) 53 m (b) 78 m
(c) 92 m (d) 120 m

50. Who is in the South-East from the person who is left from 'D'

(a) A (b) B
(c) C (d) D

51. Five cities are near to each-other. These cities are A, B, C, D, E. A is to west of B.C is to south of A. E is in North of B.D is in the east of E. In which direction is C from D?

(a) North-West (b) South-East
(c) South-West (d) None of above option

52. Vijay Kumar walks from his house in the north direction a distance of 15 km. Then he turns west direction and walks straight 10 km. At last he turns south direction and walks straight 5 km. How far and in which direction is he from his house?

(a) 10 km, East (b) 5 km, West
(c) 20 km, North (d) None of these

53. Raju is the neighbour of Ramu and he lives in South-East at the distance of 100 m. Verma is the neighbour of Raju and he lives in south- west direction and 100 m distance from Raju. Mr. Singht is the neighbour of Ramu and lives 100 m is North-East direction from Ramu. In which direction is Mr. Singh living from Mr. Verma?

(a) North-West (b) North-East
(c) South (d) South-West

Direction (54 – 56) : Each Question is based on information given below: -

Point 'D' is in west direcdtion from 'A' at the distance of 14m. Point 'B' is in south direction from 'D' also a distance of 4m. Point 'F' is situated 9 m. in south from 'D'. Point E is situated 7m in east from 'B'. Point 'C' is in north from 'E' at the distance of 4m. Point 'G' is in south from 'A' at the distance of 4 m.

54. Which is in the straight direction out of the following?

(a) D, E, A (b) E, G, C
(c) D, B, G (d) E, G, B

55. In which direction is 'A' from 'C'?

(a) East (b) West
(c) North (d) South

56. Any person from point 'F' starts walking towards north at the distance of 5 m then turns his rightwards then to whom he will reach first out of the following?

(a) G (b) D
(c) E (d) A

Directions (57 – 60): Study the informations and answer the question given below:

Eight friends, A, B, C, D, E, F, G and H are sitting in a square dining table. On one side of table, only two friends can sit. All the friends are sitting at equal distance from each other. A is facing south. 'B' and 'F' are sitting opposite side. 'F' is not facing in that direction in which 'A' and 'H' are facing, 'H' is facing west direction. 'D' and 'C' are both sitting at same side. 'C' is facing to the neigbhour of 'H'. E is F's neighbour

57. In which direction is 'G' facing?

(a) East (b) West
(c) North (d) South

58. In which direction is 'F' facing ?

(a) East (b) West
(c) North (d) South

59. In which direction is 'D' facing ?

(a) East (b) West
(c) North (d) South

60. In which direction is 'B' facing ?

(a) East (b) West
(c) North (d) South

Distinct Questions

61. Jai Prakash started walking towards south from the point A, walked 20 m and reached a point B, Again, he turned left and walked 20 m and reached a point C. Now he turned 45° anticlockwise, walked a distance of $20\sqrt{2}$ m and reached a point D. What approximately is the shortest distance between the point A and D?

(a) Can't say (b) 30 m
(c) 40 m (d) $40\sqrt{2}$ m

62. Rishu started walking towards East from a point 'X', walked 250 metres and reached the point 'P'. Again, he turned towards South, walked 50 metres and reached point Q. Again he turned towards east and walked 250 metres and reached point 'R. He again turned South, walked 50 metres and reached point S. What is the shortest distance between X and S?

(a) 509.9 metres (b) 1100 metres
(c) 561.2 metres (d) 590.9 metres

63. A and B are standing at a distance of 20 km from each other on a straight East-West road. A and B start walking simutaneously, eastwards and westwards respectively, and both cover a distance of 5 km. Then A turns to his left and walks 10 km. B turns to his right and walks 10 km at the same speed. Then both turn to their left and cover a distance of 5 km at the same speed. What will be the distance between them?

(a) 10 km (b) 5 km
(c) 20 km (d) 25 km

64. After walking 6 km, I turned right and covered a distance of 2 km, then turned left and covered a distance of 10 km. In the end, I was moving towards the north. Form which direction did I start my journey?

(a) North (b) South
(c) East (d) West

65. A boy ride his bicycle northwards, then turned left and ride one km and again turned left and ride 2 km. He found himself exactly one km west of his starting point. How far did he ride northwards initially?

(a) 1 km (b) 2 km
(c) 3 km (d) 5 km

66. Anuj started walking positioning his back towards the sun. After sometime, he turned left, then turned right and then towards the left again. In which direction is he going now?

(a) North or South (b) East or West
(c) North or West (d) South or West
(e) None of these

67. A mountaineer starts from Camp A and proceeds East to Camp B 4 km away. from camp B he proceeds to camp C, 5km to the south. from there, he returns and proceeds 12km to camp 'D' via camps B and A. How far is he away from the camp A ?

(a) 3 km (b) 6 km
(c) 5 km (d) 12 km
(e) 8 km

68. Sunil walks towards the East from point A turns righ at point B and walks the same distance as he walked towards the East. He now turns left, walks the same distance again and finally makes a left turn and stops at point C after walking the same distance. The distance between A and C is how many times as that of A and B?

(a) Can't be determined (b) Two
(c) Three (d) Four
(e) None of these

69. Shyam walked 6 metres facing towards East, then took a right turn and walked a distance of 9 metres. He then took a left turn and walked a distance of 6 metres. How far is he from the starting point ?

(a) 15 metres (b) 21 metres
(c) 18 metres (d) Can't be determined
(e) None of these

70. Ravi starts from his house and moves towards south. He walks 100 m, then turns left and walks 200 m, turns right and walks 500. How far is he from his house?

(a) $400\sqrt{5}$ m (b) 800 m
(c) $200\sqrt{10}$ m (d) $200\sqrt{2}$ m
(e) None of these

Previous year Questions (Memory Based)

1. Going 50 m to the South of her house, Radhika turns left and goes another 20 m. Then, turning to the North, she goes 30 m and then starts walking to her house, in which direction is she walking now?

(a)North-west (b) North
(c) South-east (d) East

2. I am facing South. I turn right and walk 20 m. Then I turn right again and walk 10 m. Then I turn left and walk 10 m and then turning right walk 20 m. Then I turn right again and walk 60 m. in which direction am I from the starting point?

(a)North (b) North-west
(c) East (d) North-east

3. Aditya starts from his house towards West. After walking a distance of 30 metres, he turned towards right and walked 20 metres. He then turned left and moving a distance of 10 metres, turned to his left again and walked 40 metres. He now turns to the left and walks 5 metres. Finally he turns to his left. In which direction is he walking now?

(a) North (b) South
(c) East (d) South-west

4. You go North, turn right, then right again and then go to the left. In which direction are you now?

(a) North (b) South
(c) East (d) West

5. Deepak starts walking straight towards east. After walking 75 meters, he turns to the left and walks 25 metres straight. Again the turns to the left, walks a distance of 40 meters straight, again he turns to the left and walks a distance of 25 metres. How far is he from the starting point?

(a) 25 metres (b) 50 metres
(c) 115 metres (d) None of these

6. Kunal walks 10 kilometres towards North. From there, he walks 6 kilometres towards South. Then, he walks 3 kilometres towards East. How far and in which direction is he with reference to his starting point?
(a) 5 kilometres West (b) 5 kilometres North-east
(c) 7 kilometres East (d) 7 kilometres West

7. Rohan walks a distance of 3 km towards North, then turns to his left and walks for 2 km. he again turns left and walks for 3 km. At this point he turns to his left and walks for 3 km. how many kilometers is he from the starting point?
(a) 1 km (b) 2 km
(c) 3 km (d) 5 km

8. Manick walked 40 metres towards North, took a left turn and walked 20 metres. He again took a left turn and walked 40 metres. How far and in which direction is he from the starting point?
(a) 20 metres East (b) 20 metres North
(c) 20 metres South (d) None of these

9. Namita walks 14 metres towards west, then turns to her right and walks 14 metres and then turns to her left and walks 10 metres. Again turning to her left she walks 14 metres. What is the shortest distance (in metres) between her starting point and the present position?
(a)10 (b) 24
(c) 28 (d) 38

10. A man leaves for his office from his house. He walks towards East. After moving a distance of 20 m, he turns South and walks 10 m. then he walks 35 m towards the West and further 5 m towards the North. He then turns towards East and Walks 15 m. What is the straight distance (in Metres) between his initial and final positions?
(a) 0 (b) 5
(c) 10 (d) Cannot be determined

11. Radha moves towards South-east a distance of 7 km, then she moves towards West and travels a distance of 14 m. From here, she moves towards North-west a distance of 7 m and finally she moves a distance of 4 m towards East and stood at that point. How far is the starting point from where she stood?
(a) 3 m (b) 4 m
(c) 10 m (d) 11 m

12. Amit walked 30 metres towards East, took a right turn and walked 40 metres. Then he took a left turn and walked 30 metres. In which direction is he now from the starting point?
(a) North-east (b) East
(c) South-east (d) South

13. A person starts from a point A and travels 3 km eastwards to B and then turns left and travels thrice that distance to reach C. He again turns left and travels five times the distance he covered between A and B and reaches his destination D. the shortest distance between the starting point and the destination is
(a)12 km (b) 15 km
(c) 16 km (d) 18 km

14. A girl leaves from her home. She first walks 30 metres in North-west direction and then 30 metres in South-west direction. Next, she walks 30 metres in South-east direction. Finally, she turns towards her house. In which direction is she moving?
(a)North-east (b) North-west
(c) South-east (d) South-west

15. Sanjeev walks 10 metres towards the South. Turning to the left, he walks 20 metres and then moves to his right. After moving a distance of 20 metres, he turns to the right and walks 20 metres. Finally, he turns to the right and moves a distance of 10 metres. How far and in which direction is he from the starting point?
(a)10 metres North
(b) 20 metres South
(c) 20 metres North
(d) 10 metres South

16. Kashish goes 30 metres North, then turns right and walks 40 metres, then again turns right and walks 20 metres, then again turns right and walks 40 metres. How many metres is he from his original position?
(a)0 (b) 10
(c) 20 (d) 40

17. A man walks 30 metres towards South. Then, turning to his right, he walks 30 metres. Then, turning to his left, he walks 20 metres. Again, he turns to his left and walks 30 metres. How far is he from his initial position?
(a)20 metres (b) 30 metres
(c) 60 metres (d) None of these

18. Rohit walked 25 metres towards South. Then he turned to his left and walked 20 metres. He then turned to his left and walked 25 metres. He again turned to his right and walked 15 metres. At what distance is he from the starting point and in which direction?
(a) 35 metres East (b) 35 metres North
(c) 40 metres East (d) 60 metres East

19. Starting from a point P, Sachin walked 20 metres towards South. He turned left and walked 30 metres. He then turned left and walked 20 metres. He again turned left and walked 40 metres and reached a point Q. How far and in which direction is the point Q from the point P?

(a) 20 metres West (b) 10 metres East
(c) 10 metres West (d) 10 metres North

20. Ramakant walks northwards. After a while, he turns to his right and a little further to his left. Finally, after walking a distance of one kilometer, he turns to his left again. In which direction is he moving now?
(a) North (b) South
(c) East (d) West

21. A man is facing south. He turns 135° in the anti clockwise direction and then 180° in the clockwise direction. Which direction is he facing now?
(a)North-east (b) North-west
(c) South-east (d) South-west

22. A man is facing north-west. He turns 90° in the clockwise direction and then 135° in anti clockwise direction. Which direction is he facing now?
(a)East (b) West
(c) North (d) South

23. A man is facing towards west and turns through 45° clockwise, again 180° clockwise and then turns through 270° anti clockwise. In which direction is he facing now?
(a)West (b) North-east
(c) South (d) South-west

24. A river flows west to east and on the way turns left and goes in a semi-circle round a hillock, and then turns left at right angles. In which direction is the river finally flowing?
(a) West (b) East
(c) North (d) Sourh

25. I am Standing at the centre of a circular field. I go down south to the edge of the field and then turning left I walk along the boundary of the field equal to three-eights of its length. Then I turn left by 45° and go right across to the opposite point to the boundary. In which direction am I from the starting point?
(a) North-west (b) North
(c) South-west (d) West

26. I am facing east. I turn 100° in the clockwise direction and then 145° in the anti clockwise direction. Which direction am I facing now?
(a) East (b) North-east
(c) North (d) South-west

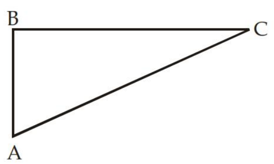
27. A rat runs 20´ towards East and turns to right, runs 10´ and turns to right, runs 9´ and again turns to left, runs 5´ and then turns to left, runs 12´ and finally turns to left and runs 6´. Now, which direction is the rat facing?
(a) East (b) West
(c) North (d) South

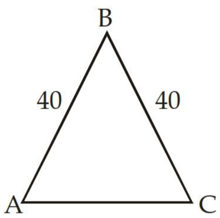
28. Maya starts at point T, walks straight towards North to point U which is 4 ft away. She truns left at 90° and walks 1 ft to Q, turns left at 90° and goes to V, who is 1 ft away and once again turns 90° right and goes to R, 3 ft away. What is the distance between T and R?
(a) 4 ft (b) 5 ft
(c) 7 ft (d) 8 ft

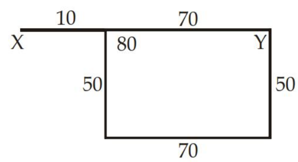
29. A villager went to meet his uncle in another village situated 5 km away in the North-east direction of his own village. From there he came to meet his father-in-law living in a village situated 4 km in the south of his uncle's village. How far away and in what direction is he now?
(a)3 km in the North (b) 3 km in the East
(c) 4 km in the East (d) 4 km in the West

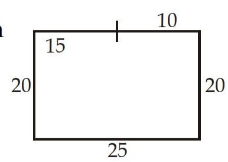
30. Aditya walked 15 m. towards south took a right turn and walked 3 m. He took a right turn again and walked 15 m before stopping. Which direction did he face.
(a) East (b) West
(c) North (d) South

Practice Set Solutions

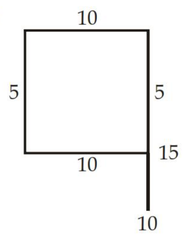
1. **(d);**


2. **(a);**


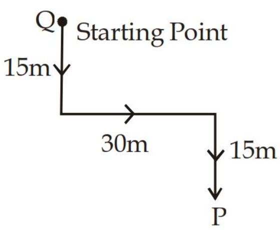
3. **(a);**


4. **(b);** 10 km


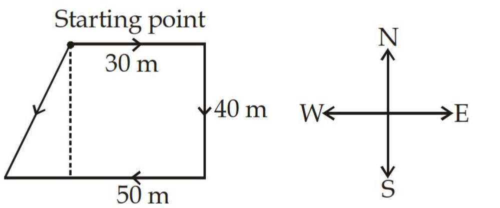
5. **(c);**



6. **(b);** Obviously, Z is facing south

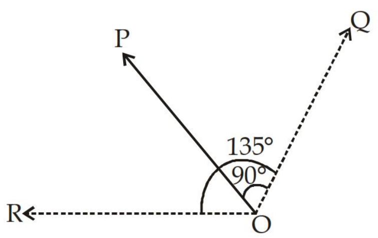


7. **(c);**

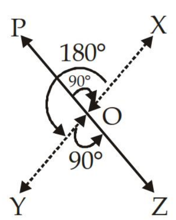


Clearly, Alok is in South-West direction from starting point.

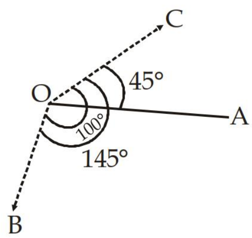
8. **(b);** As shown in Fig. the man initially faces in the direction OP. On moving 90° clockwise the man faces in the direction OQ. On further moving 135° anticlockwise, he faces in the direction OR, which is West.



9. **(d);** As shown in Fig. the man intially faces in the direction OP. On moving 90° clockwise, he faces in the direction OX. On further moving 180° anticlockwise, he faces in the direction OY. Finmally, on moving 90° anticlockwise, he faces in the direction OZ, which is South-east.



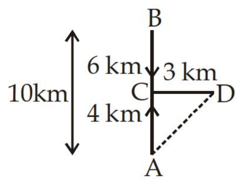
10. **(b);** As shown in Fig. the man initially faces towards eact i.e., in the firection OA. On moving 100° clockwise he faces in the directin OB. On further moving 145°clockwise, he faces in the dirction OC, Claearely, OC makes an angle of (145°–100°) i.e. 45° with OA and as such points in the direction North-east.



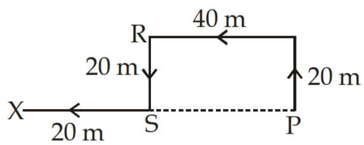
11. **(b);** The movements of Kishankant are as shown in Fig. (A to B, B to C and C to D).
AC = (AB-BC) =(10–6) km = 4 km
Kishankant's distance from startimg point A

$= AD = \sqrt{AC^2 + CD^2} = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$ km

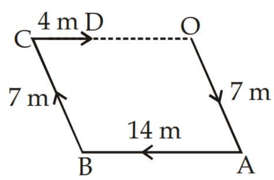
So Kishankant is 5 km to the North-east of his starting point.



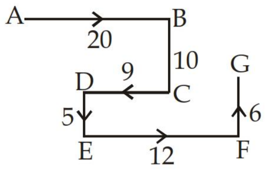
12. **(d);** The movements of Gaurav are as shown in Fig. Clearly, Gaurav's distance from his initial position
P = PX = (PS + SX)
= (QR + SX) = (40 + 20)m = 60.



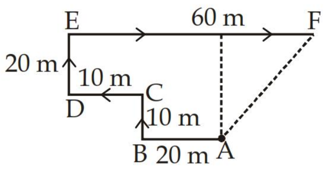
13. **(c);** The movements of Radha are as shown in Fig. Clearly, Radha's distance from the staring point
O = OD = (OC–CD) = (AB–CD) = (14–4) m = 10 m



14. **(c);** The movements of the rat from A to G are as shown in Fig.
Clearly, it is finally walking in the direction FG, ie North.



15. (d); The movements of the person are form A to F, as shown in Fig. Clearly, the final position is F which is to the North-east of the starting point A

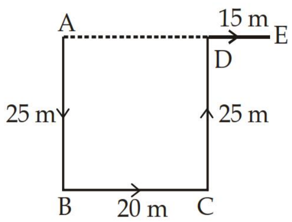


16. (a); The movements of Rohit are as shown in fig.

∴ Rohit's distance from starting point A =

AE = AD + DE = BC + DE = (20 + 15)m = 35 m

Also, E is to the East of A



17. (c); The movements of Sachin are as shown in Fig.

(P to B, B to C, C to D and D to Q)

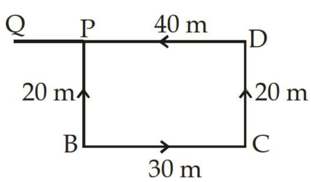
Clearly, distance of Q from P

= PQ = (DQ - PD) = (DQ - BC)

= (40 - 30) m = 10 m

Also, Q is to the West of P.

∴ Q is 10 m West of P.



18. (c); The movements of the man are as shown in Fig.

(A to B, B to C, C to D and D to E)

Clearly, DF = BC = 5 km

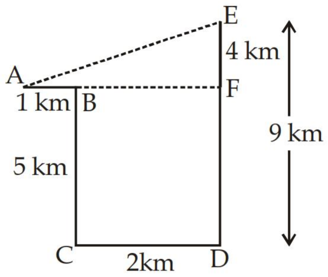
EF= (DE- DF) = (9-5) km = 4 km

BF = CD = 2 km

AF = AB + BF = AB + CD = (1 + 2) km = 3 km

∴ Man's distance from starting point A = AE

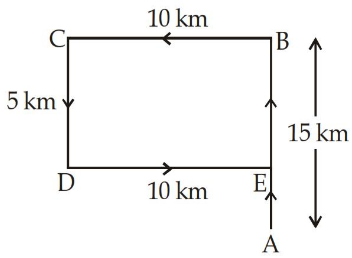
$= \sqrt{AF^2 + EF^2} = \sqrt{3^2 + 4^2} = \sqrt{25} = 5\,\text{km}$



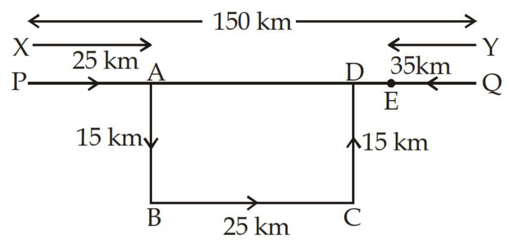
19. (c); The movements of Lokesh are as shown in Fig.

(A to B, B to C, C to D and D to E).

Clearly, his final position is E which is to the North of his house at A.



20. (a); Let X and Y be two buses.



Bus X travels along the path PA, AB, BC, CD.

Now, AD = BC =25 km

So, PD = PA + AD = 50 km

Bus Y travels 35 km upto E

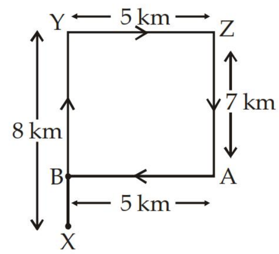
∴ Distance between two buses = PQ - (PD + QE)

= {150 - (50+35)} = 65 km

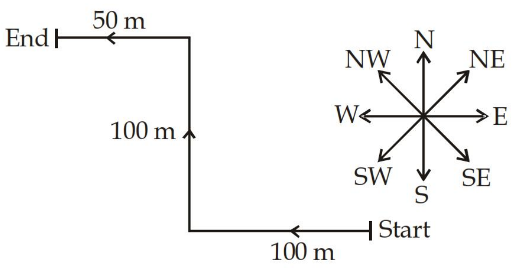
21. (a); Clearly, the route of Sohan is as shown in the diagram given below :

Here, XB = XY – YB = XY – AZ (∴ YB = AZ)

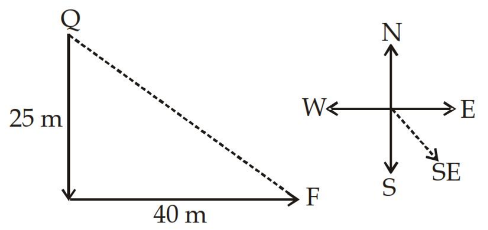
= 8 km – 7 km = 1 km



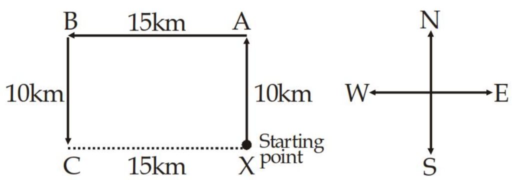
22. (d); From the figure given below, Raj is finally in the North-West direction from the starting point.



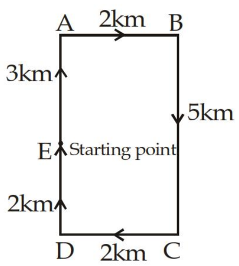
23. (d); From the figure, it is clear that Alok is in the South-East direction with reference to starting point Q.



24. (d);

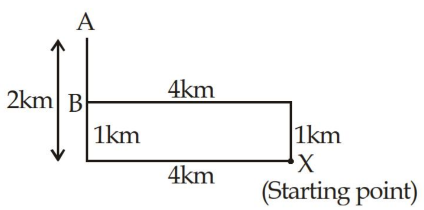


25. (a);



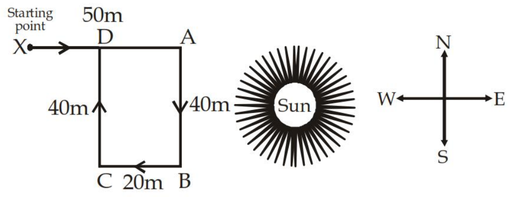
Finally he is facing North direction.

26. (a);



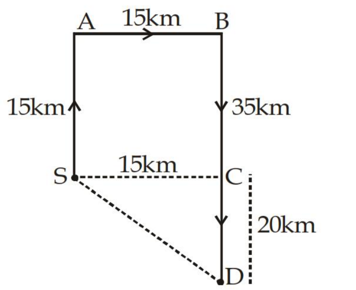
Clearly the distance between A and B = 1km

27. (b);



XD = XA – AD = 50 – 20 = 30m

28. (b);



Let his starting point is S.

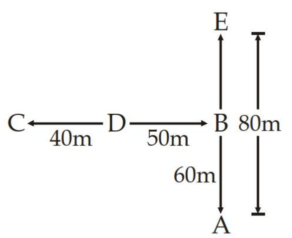
AB = SC =15 m, BC = AS = 15km

BD - BC = 35 - 15 = 20 km

SD = $\sqrt{SC^2 + CD^2} = \sqrt{15^2 + 20^2}$

$= \sqrt{225 + 400} = \sqrt{625} = 25\text{km}$

(29 – 31):



29. (d); Total distance walked by the boy.

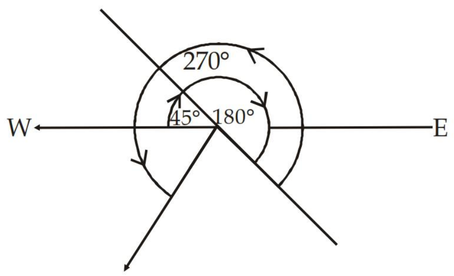
= 40 + 50 + 60 + 80 = 230 m

30. (c); CE = $\sqrt{BC^2 + BE^2} = \sqrt{90^2 + 20^2}$

$= \sqrt{8100 + 400} = \sqrt{8500} = 92.19\text{m}$

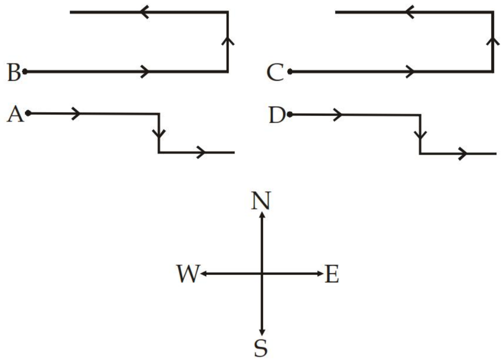
31. (a);

32. **(d);**

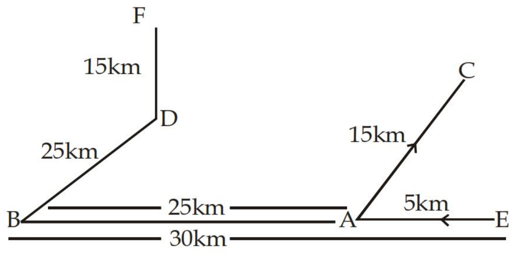


So, finally he is facing South-West direction.

33. (a);



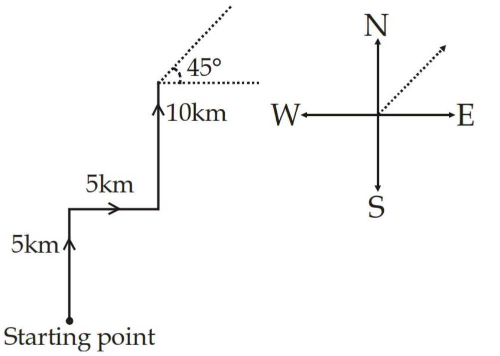
34. (a); The diagram of check post according to direction.



EF = BE + BD + DF = 30 + 25 +15 = 70km

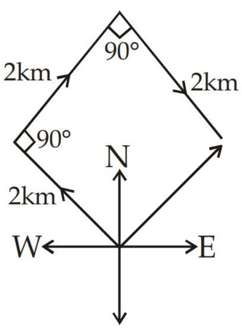
35. (b); B is located to south-west of D.

36. (c);

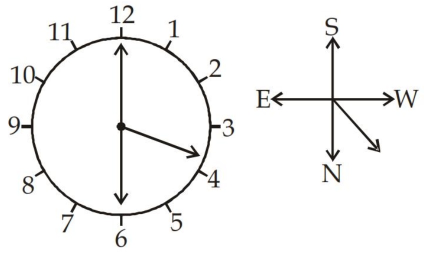


Finally Vishesh is facing North-East direction

37. (b);

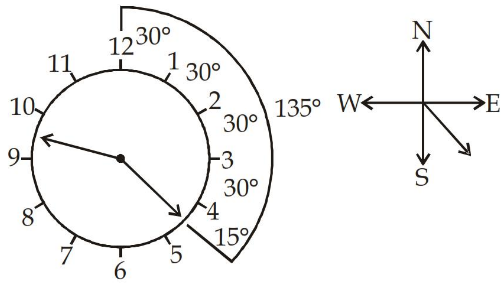


38. (a);



Obviously, an hour hand points towards North-West.

39. (a);



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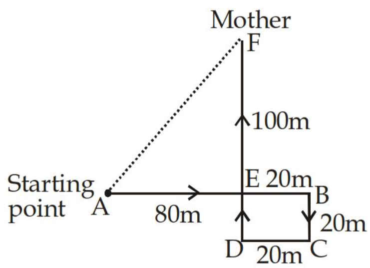
40. (c);



AE = AB - EB

= 80 – 20 = 60m

EF = DF - DE

= 100 - 20 = 80m

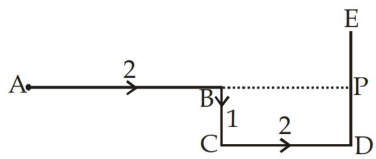
Now in Δ AEF,

AF = $\sqrt{(AE)^2 + (EF)^2}$

$= \sqrt{60^2 + 80^2}$

$= \sqrt{3600 + 6400} = \sqrt{10000} = 100\text{m}$

41. (c);

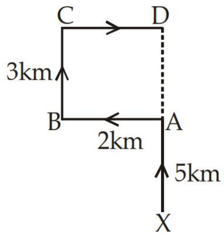


AP = 2 + 2 = 4 km

PE = 4 – 1 = 3km

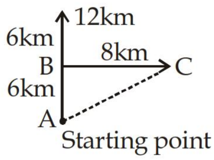
$\Rightarrow$ AE = $\sqrt{16+9} = 5\text{km}$

42. (b);



XD = XA + AD = 5 + 3 = 8 km

43. (c);

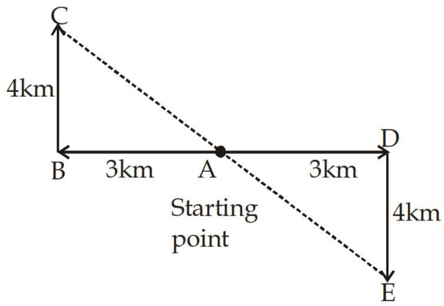


$AC = \sqrt{(AB)^2 + (BC)^2} = \sqrt{(6)^2 + (8)^2}$

$= \sqrt{36 + 64} = \sqrt{100} = 10km$

and the direction is North-East

44. (d);



In Δ ABC, $AC = \sqrt{(AB)^2 + (BC)^2}$

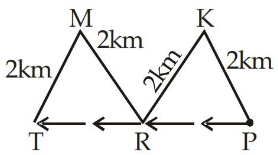
$= \sqrt{(3)^2 + (4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5km$

In Δ ADE, $AE = \sqrt{(AD)^2 + (DE)^2}$

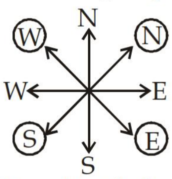
$= \sqrt{(3)^2 + (4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5km$

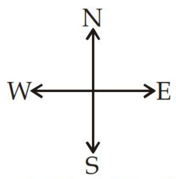
Hence, CE = 5 + 5 = 10km

45. (b); West



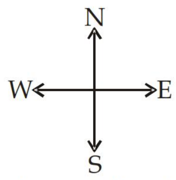
46. (a); According to question



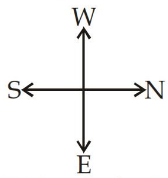


Now North direction is West-North or North-West.

47. (a);

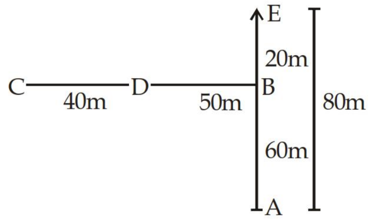


changed direction



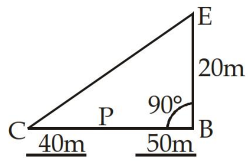
Oviously, he is going in West direction.

48. (d);



Required distance = 40 + 50 + 60 + 80 = 230 m

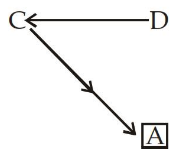
49. (c);



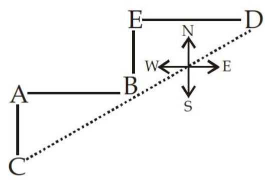
$EC = \sqrt{(CB)^2 + (BE)^2} = \sqrt{(90)^2 + (20)^2}$

$= \sqrt{8100 + 400} = \sqrt{8500} = 92m$ (near about)

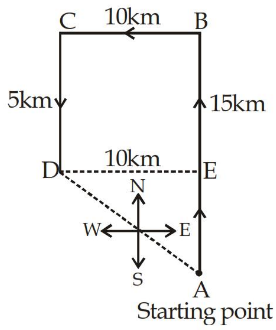
50. (a);



51. (c);



52. (d);



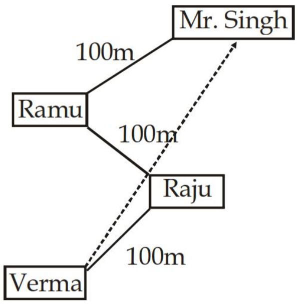
DE = CB =10km
AE = AB - BE = 15 – 5 = 10km
(CD = BE)

$AD = \sqrt{DE^2 + AE^2} = \sqrt{(10)^2 + (10)^2}$

$= \sqrt{100+100} = \sqrt{200} = \sqrt{100 \times 2} = 10\sqrt{2}$

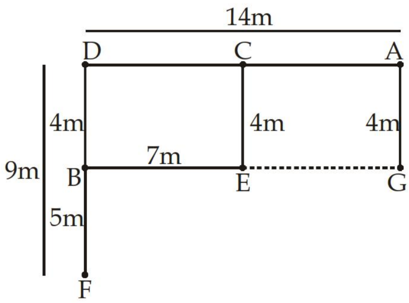
Thus it's $10\sqrt{2}$ km in North-West direction.

53. (b);



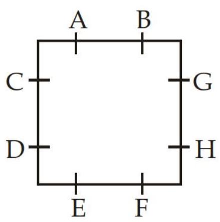
Mr. Singh is in North-East direction from Verma.

(54 – 56):



54. (d); From given option E,B, G are in a straight line
55. (a); 'A' is in the East of 'C'.
56. (c); Surely, he will reach at point 'E' first.

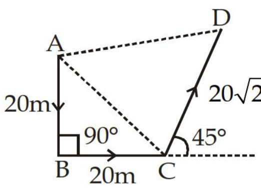
(57 – 60):



57. (b); **58. (c);** **59. (a);** **60. (d);**

Distinct Solutions

61. (c);



Here, $AC = \sqrt{20^2 + 20^2} = 20\sqrt{2}$ m

Since AB = BC

$\therefore \angle ACB = \angle BAC = 45°$

$\therefore \angle ACD = 90°$

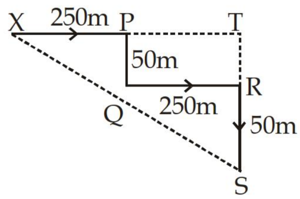
Now,

$AD^2 = AC^2 + CD^2 = \left(20\sqrt{2}\right)^2 + \left(20\sqrt{2}\right)^2$

= 800 + 800 = 1600

$\therefore AD = \sqrt{1600} = 40$ m

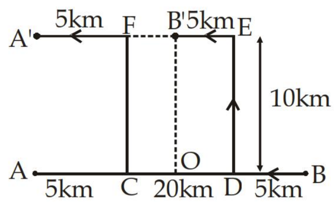
62. (a);



$XS = \sqrt{XT^2 + TS^2} = \sqrt{500^2 + 100^2}$

$= \sqrt{250000 + 10000} = 509.9$ metres.

63. **(a);**



The above fig. shows the movements of A and B.

We have to find A'B'

A'B' = A'F+FB'

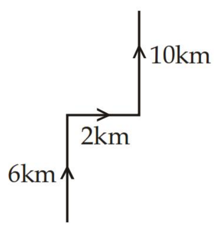
FB' = AB – (AC + BD + OD)

= [20 –(5 + 5 + 5) =] 5km

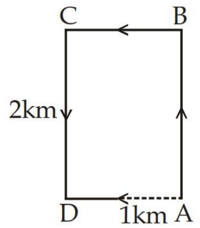
$\therefore$ A'B' = [5 + 5] = 10 km

[$\because$ A'F = 5 km, from the fig.]

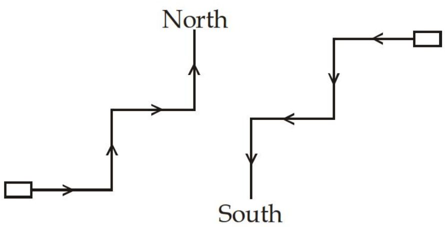
64. **(b);** Clearly, the route is as shown in the diagram. Thus, the man started his journey from the South and moved northwards.



65. (b); Clearly, the boy ride from A to B, then to C and finally upto D Since D lies to the west of A, so requied distance = AB = CD = 2 km.

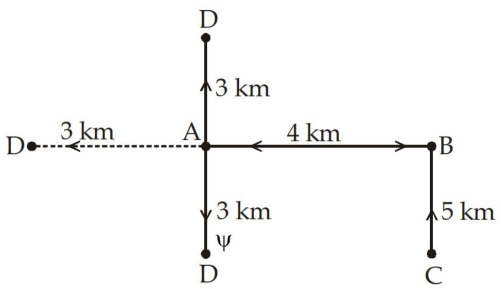


66. (a); Clearly, there are two possible movement of Anuj as shown below :

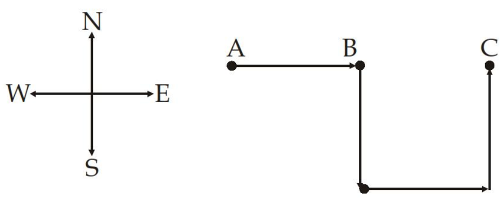


Thus, Anuj is finally moving towards either North or South.

67. (a); Clearly, mountaineer is [(12- (4 + 5) =]3 km away from the point A

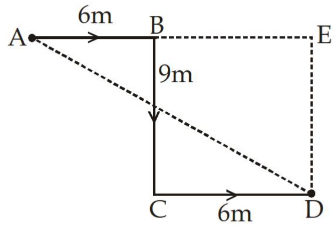


68. (b); Movements of Sunil are as shown in figure.



Hence, AC = AB + BC = 2AB

69. (a); Movements of Shyam are as shown in the figure given below:



A = Starting point

D = End point

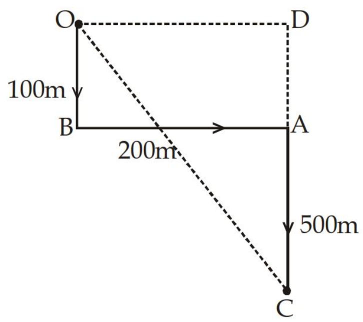
$AD = \sqrt{AE^2 + ED^2}$; Here AE = AB + BE (=CD)

= (6 + 6) = 12m

& ED = BC = 9 m

$\therefore$ $AD = \sqrt{12^2 + 9^2} = \sqrt{144 + 81} = \sqrt{225} = 15m$

70. (c); Ravi starts from O and finished at C.



$OC = \sqrt{OD^2 + CD^2} = \sqrt{OD^2 + (DA + AC)^2}$

$= \sqrt{(200)^2 + (100 + 500)^2}\,m$

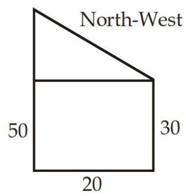
$= \sqrt{200 \times 200 + 600 \times 600}$ m

$= 100\sqrt{4 + 36}$ m $= 100\sqrt{40}$m $= 200\sqrt{10}$m

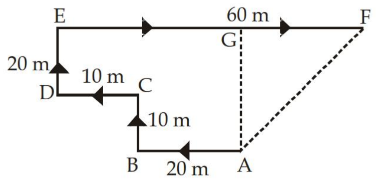
[$\because$ OD = AB = 200 m ; DA = OB = 100 m and AC = 500 m]

Previous year Solutions (Memory Based)

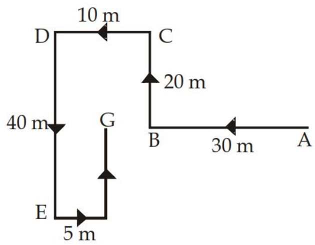
1. (a);



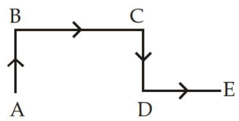
2. (d); The movements of the person are from A to F, as shown in figure. Clearly, the final position is F which is to the North-east of the starting point.



3. (a); The movements of Aditya are shown in figure from A to G. Clearly, Gopal is finally walking in the direction FG i.e., North.

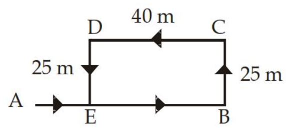


4. (c); The movements indicated are as shown in figure. Thus, the final movements in the direction indicated by DE, which is east.



5. (d); The movements is shown in figure— Clearly, EB = DC = 40 m.

So, distance from the starting point A = (AB – EB) = (75 – 40) m = 35 m.

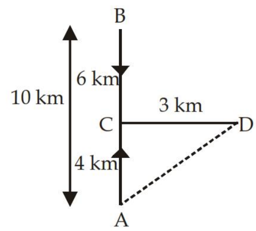


6. (b); The movements is as shown in figure.

AC = (AB – BC) = (10 – 6) km = 4 km. Clearly, D is to the North-east of A.

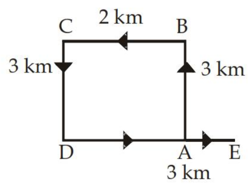
So, Kunal's distance from starting point A

$= AD = \sqrt{AC^2 + CD^2} = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$ km. So, it is 5 km to the North-east of his starting point

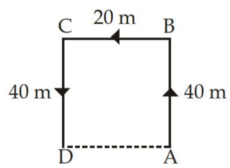


7. (a); The movements of Rohan are as shown in figure (A to B, B to C, C to D and D to E). Clearly, AD = BC = 2 km

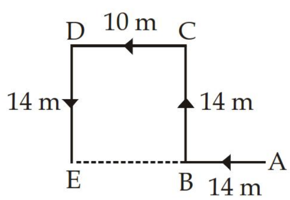
So, required distance = AE = (DE – AD) = (3 – 2) = 1 km.



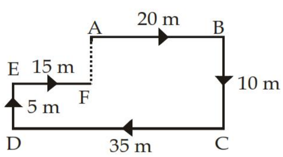
8. (d); The movements of Manick are as shown in figure (A to B, B to C and C to D). Clearly, ABCD is a rectangle and so AD = BC = 20 m. Thus, D is 20 m to the west of A.



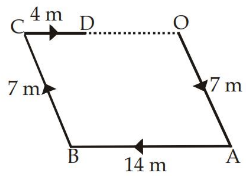
9. (b); The movements of Namita are shown in figure (A to B, B to C, C to D and D to E). Clearly, Namita's distance from his initial position = AE = (AB + BE) = (AB + CD) = (14 + 10) m = 24 m.



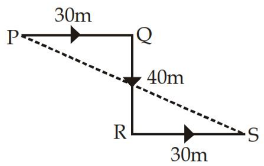
10. (b); Clearly, DC = AB + FE. So, F is in the line with A. Also, AF = (BC – DE) = 5 m. So, the man is 5 metres away from his initial position.



11. (c); The movements of Radha are as shown in figure. Clearly, Radha's distance from the starting point O = OD = (OC – CD) = (AB – CD) = (14 – 4) m = 10 m.



12. (c); The movements of Amit are shown in figure (P to Q, Q to R and R to S). Clearly, his final position is S which is to the South-east of the starting point P.



13. (b); The movements of the person are as shown in figure.

Clearly, AB = 3 km; BC = 3AB = (3 × 3) km = 9 km; CD = 5 AB = (5 × 3) km = 15 km.

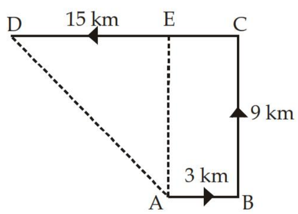
Draw $AE \parallel CD$. Then, CE = AB = 3 km and AE = BC = 9 km.

DE = (CD – CE) = (15 – 3) km = 12 km. In “AED,

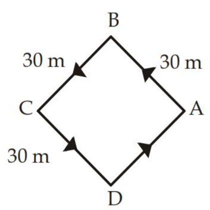
$AD^2 = AE^2 + DE^2$; AD = $\sqrt{9^2 + 12^2}$

= $\sqrt{225}$ = 15 km. So, required distance

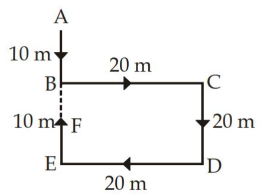
= AD = 15 km.



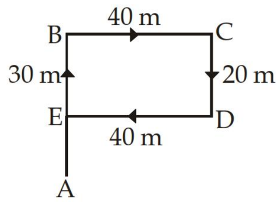
14. (a); The movements of the girl are as shown in figure (A to B, B to C, C to D, D to A). Clearly, she is finally moving in the direction DA i.e., North-east.



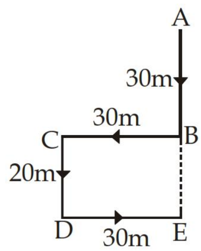
15. (b); The movements of Sanjeev from A to F are shown in figure. Clearly, Sanjeev's distance from starting point A = AF = (AB + BF) = AB + (BE – EF) = AB + (CD – EF) = [10 + (20 – 10)] m = (10 + 10) m = 20 m. Also, F lies to the South of A. So, Sanjeev is 20 metres to the south of his starting point.



16. (b); The movements of Kashish are as shown in figure (A to B, B to C, C to D, D to E). So, Kashish's distance from his original position A = AE = (AB – BE) = (AB – CD) = (30 – 20) m = 10 m.

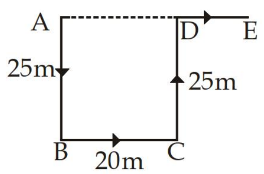


17. (d); The movements of the man are as shown in figure. So, Man's distance from initial position A = AE = (AB + BE) = (AB + CD) = (30 + 20) m = 50 m.

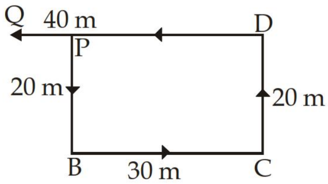


18. (a); The movements of Rohit are as shown in the figure.

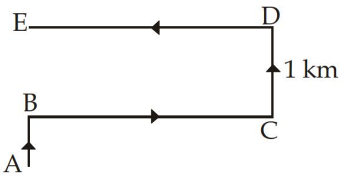
So, Rohit's distance from starting point A = AE = (AD + DE) = (BC + DE) = (20 + 15) m = 35 m. Also, E is to the East of A.



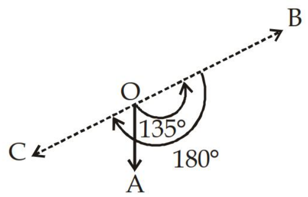
19. (c); The movements of Sachin are as shown in figure (P to B, B to C, C to D and D to Q). Clearly, distance of Q from P = PQ = (DQ – PD) = (DQ – BC) = (40 – 30) m = 10 m. Also, Q is to the West of P. So, Q is 10 m West of P.



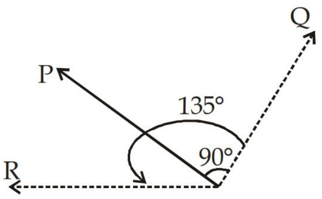
20. (d); The movements of Ramakant are as shown in figure. Clearly, he is finally walking in the direction DE i.e., West.



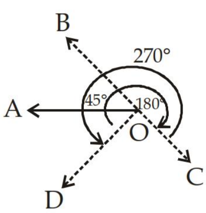
21. (d); As shown in figure , the man initially faces in the direction OA. On moving 135^0 anti-clockwise, he faces in the direction OB. On further moving 180^0 clockwise, he faces in the direction OC, which is South-west.



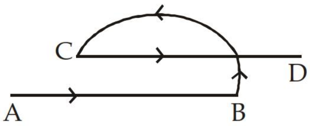
22. (b); As shown in figure, the man initially faces in the direction OP. on moving 90^0 clockwise, the man faces in the direction OQ. On further moving 135^0 anti clockwise, he faces in the direction OR, which is West.



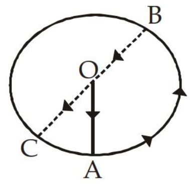
23. (d);



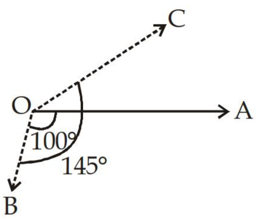
24. (b); As shown in figure, the river flows eastwards from A towards B, turns left and follows a semi-circular path to reach C where it turns left and flows east wards towards D.



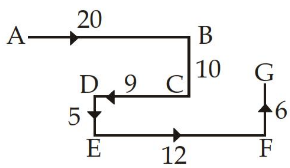
25. (c); The movements are as indicated in figure (O to A, A to B and B to C). Clearly, C lies to the South-west of O.



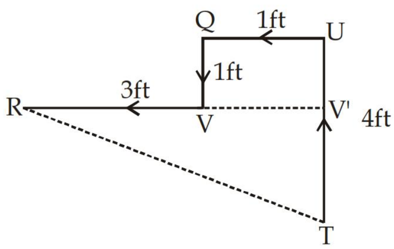
26. (b); As shown in figure, the man initially faces towards east i.e., in the direction OA. On moving 100^0 clockwise, he faces in the direction OB. On further moving 145^0 clockwise, he faces in the direction OC. Clearly, OC makes an angle of (145^0 - 100^0) i.e., 45^0 with OA and as such points in the direction North-east.



27. (c); The movements of rat from A to G are as shown in figure. Clearly, it is finally walking in the direction FG i.e., North.

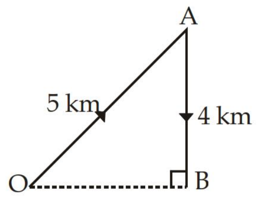


28. (b); The movements of Maya from T to R are as shown in figure.



RT = $\sqrt{(RV')^2+(TV')^2} = \sqrt{16+9} = 5ft$

29. (b); The villager moves from his village at O to his uncle's village at A and thereon to his father-in-law's village to B. Clearly, " OBA is right-angled at B. So, $OA^2 = OB^2 + AB^2$; $OB^2 = OA^2 - AB^2$; $OB = \sqrt{25-16} = \sqrt{9} = 3$ km. Thus, B is 3 km to the east of his initial position O.



30. (c);